

Sources for model validation, membership and allocation

Mapping evidence retained from 3 October 2026; corrected-model diagnostics and source corrections verified 8 October 2026.

Source	Reference	Link	What the source supports
CompositionSearch	Primary composition searches, decisions and bounded access failures	Open source	Actual final query strings and provider outputs; applicable observed composition not established, so modeled proxy allocations remain explicitly assumed.
Focal	Chiaverano et al. (2018), Northern Humboldt ecological network	Open source	Focal inheritance, domain/period, added jellyfish/hurles/eggs, source-version qualifications, DOI 10.1016/j.pocean.2018.04.009.
Native	Original revised/final supplement, Table A: resolved stock parameters and fleet removals	Open source	Original Table A supplies published landings/discards and allocation proxies. These source proxies are retained; accepted sardine removals/EE corrections are separate.
FreshTaxonomy	Independent fresh group definitions and evidence scopes	Open source	Exact stock definitions; inherited enumerated examples distinguished from parameter analogues and unsupported exhaustive lists.
FreshProxy	Independent source catch and biomass vector with exact cell locators	Open source	Every candidate raw value and genuine zero, source basis and fallback inputs.
Tam2008	Tam et al. (2008), inherited Northern Humboldt model methods, pp.353–354	Open source	Explicit species examples, three Peruvian hake intervals, plankton size classes; parameter analogues are not source-membership observations.
Regional	LME 013 regional Catch, Classic PPR and NPP tables, read-only snapshot	Open source	All exact reporting labels and functional/common names, all bases/years, saved independent classic TL/SPPR and NPP. Active selection was not changed.
SAU_methods	Sea Around Us, catch reconstruction and allocation methods	Open source	Commercial/functional reporting groups differ from taxa and maximum size is not caught-size mass composition; page opened 2026-10-03.
WoRMS	WoRMS/FishBase/MerlucciaBase authority records retained from 30 September 2026	Open source	Taxonomic identity, accepted name, family/order, marine/freshwater status and authority IDs for all available exact catch names. These records are reused taxonomy evidence, not prior mapping assignments.
WoRMS_hake	WoRMS Merluccidae taxonomic list and accepted Peruvian hake identity	Open source	Merluccius gayi peruanus accepted as M. peruanus, M. gayi and M. australis retained distinct. Search result read; full endpoint fetch failed 2026-10-03.
OBIS_Loligo	OBIS, Loligo gahi taxon entry	Open source	Accepted Doryteuthis gahi identity for source Loligo gahi; full entry opened 2026-10-03.
IMARPE_hake2019	IMARPE 2019 Scientific and Technical Yearbook, p.38, hake fishery size structure	Open source	Primary indexed passage describes industrial catch lengths, not matching all-region caught-mass bins for exact model intervals. Full PDF retrieval failed; no fractions adopted.
IMARPE_sciainds	IMARPE 2016 Vol.43 No.2, p.158, coastal fishery species and caught lengths	Open source	Regional small scientiaids and historical family nomenclature; primary indexed table only. A single coastal fishery does not supply whole-LME residual-label proportions.
IMARPE_congrito	IMARPE 1995–1996 commercial species list and later demersal records	Open source	Primary indexed commercial list names Congrito = Genypterus maculatus; later records distinguish Ophichthus angula. Supports generic-pool interpretation, not focal explicit membership.
IMARPE_murinda	IMARPE 2025 pelagic survey, p.34, Grimothea monodon in midwater catch	Open source	Primary indexed pelagic occurrence and meaningful squid-lobster analogue; survey catches do not determine broad regional commercial residual composition.
IMARPE_inverts	IMARPE public fishery activity reporting service and 2013 marine invertebrate landings	Open source	Regional fisheries report scallop/small/octopus/squid/crab resources separately; national totals do not establish catch composition of unidentified SAU labels.
FAO_sharks	FAO shark habitat/species catalogue and Galeorhinus fishery review	Open source	Mixed demersal/pelagic Galeorhinus ecology; primary FAO catalogue and regional records inform elasmobranch guilds. No compatible regional observed habitat-pool catch fractions were adopted.
FishBase_roughy	FishBase orange roughy species summary	Open source	Deep bathypelagic habitat (typically 400–800m) differs from the represented coastal guild; indexed primary authority page read 2026-10-03.
NIWA_galaxias	NIWA, new science on inanga	Open source	Marine larvae and freshwater adult life establish a true life-history mismatch, rather than following the inconsistent provider large-reef tag.
Candidate audit	Fresh extraction, diagnostics and candidate calculations	Open source	Source identity, scientific restrictions, complete run evidence
Regional input	LME_013.xlsx Catch, Classic PPR / Taxa, NPP	Open source	2019 landings and independent saved classic coefficients; NPP series in tonnes C/year
Validation	Northern Humboldt candidate validation DOCX	Open source	Candidate review awaiting researcher decision
AcceptedCorrections	Source-backed jellyfish reconstruction and article-priority sardine correction, 8 October 2026	Model notes	Adopted diet 68% meso / 17% macro / 15% eggs; sardine landings 1.4, source discards retained, EE recalculated. Original author-source contradictions remain documented.